

WILDFIRE PREDICTION USING DATASCIENCE AND MACHINE LEARNING

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Abstract—Wildfires are a risk to both wildlife and the environment, and they also represent a substantial danger to human life, leading to extensive economic and ecological damage. An early warning of wildfires can reduce these losses by allowing a window of time for preventive measures. This project is concerned with designing a Wildfire Predictive Model based on Data Science and (ML) approaches to predict wildfires depending on weather conditions. The data for this project have been obtained from UCI Machine Learning Repository and are wildfire data from two regions of Algerian: Belief and Side Blabbed between June 2012 and September 2012. The project consists of gathering information, preparing it, conducting exploratory data analysis (EDA), and creating models. In the classification task of wildfire occurrence predictions, the Random Forest Classifier achieved 88% accuracy. In Fire Weather Index (FWI) predictions (regression task), an R2 score of 0.92 was attained via the Random Forest Regressor. MongoDB Compass was used to store the data. The project demonstrates how machine learning algorithms are used in real-world environmental issues. The project can be extended by adding real-time weather data, issuing alerts to forest departments, and expanding the geographical area of the model. The solution has the potential to be a significant contributor to early wildfire detection and prevention and to conserving natural resources and human lives.

Keywords—support vector machine, satellite imagery, early warning system, fire detection

I.INTRODUCTION

Wildfires are most severe natural disasters that cause widespread disaster to the environment, wildlife, and human lives. Large areas of forest land are being burnt each year, resulting in a loss of biodiversity, natural resources, and human habitats. Advanced wildfire predictions have the capability to minimize these losses by allowing forest officials and authorities to take preventive measures and respond in a timely manner. This project's primary goal is to create a machine learning and data science-based model for forest fire prediction that can determine whether or not a

wildfire is likely to start based on meteorological variables including temperature, humidity, wind speed, and rainfall. This model can be used to lower the danger of forest fire outbreaks and preserve forest areas by using historical data on wildfires and weather. A dataset from the UCI Machine Learning Repository has been used for this project that includes forest fire data collected in two regions in Algeria, i.e., Bejaia and Sidi BelAbbes, from June 2012 to September 2012. The data includes several weather parameters such as Temperature, Rainfall, Relative Humidity, Wind Speed, and Fire Weather Index (FWI) that are significant contributing factors in forest fire cases. The project has many key phases such as data collection, data preprocessing, data analysis, and model building. A range of Machine Learning algorithms has been utilized to determine the likelihood of a fire occurring based on specific weather conditions. In this project, classification tasks have incorporated Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbour (KNN), and XGBoost. Furthermore, various regression algorithms have been utilized to forecast the Fire Weather Index (FWI), such as Linear Regression, Random Forest Regressor, and Support Vector Regressor (SVR). Upon evaluating the models' performance, it was observed that both the Random Forest Classifier and the Random Forest Regressor exhibited the greatest accuracy and efficiency. This exemplifies the effective application of Data Science and Machine Learning in tackling real-world issues, such as the prevention of wildfires. By forecasting potential forest fires through the examination of weather patterns, forest management teams and disaster response agencies can implement proactive strategies in advance. In the future, this model can be enhanced by combining real-time weather information, increasing accuracy in forecasts, and increasing geographical coverage.

II.PROBLEM STATEMENT

Wildfires are unpredictable natural disasters that cause extensive damage to forests, wildlife, human settlements, and infrastructure. Due to rising global temperatures and climate change, their frequency and intensity have significantly increased. Traditional prediction methods, which rely on static or empirical models, lack the accuracy required for

early detection and rapid response. Accurate wildfire prediction is challenging because of the complex interactions among meteorological factors, vegetation, terrain, and human activities. Unpredictable weather patterns, inconsistent datasets, and limited scalability of region-specific models further reduce prediction reliability. To address these issues, a data-driven approach using machine learning is essential. This project aims to develop a machine learning-based wildfire prediction model that classifies fire-prone areas (fire vs. no fire) and predicts the Fire Weather Index (FWI) using meteorological parameters. The model leverages historical data, predictive analytics, and cloud-based storage (MongoDB) for efficient data processing. This approach enhances early wildfire detection and supports effective disaster management and prevention strategies.

III.LITERATURE SURVEY

Wildfire prediction has been an important area for research field due to its devastating impact on the environment and human life. Numerous studies have been conducted using traditional statistical models, remote sensing techniques, and advanced machine learning algorithms to predict wildfire occurrences. One of the earliest approaches involved the use of meteorological data and fire factors, such as the (FWI), to assess wildfire risks. Studies showed that temperature and humidity and the wind speed, and precipitation played critical roles in fire outbreaks. With advancements in remote sensing and GIS (Geographic Information Systems), researchers started the usage of the satellite imagery to detect and monitor fires. NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite) provide real-time fire location data, which significantly improved prediction models. More recently, ML and DL have completely changes the wildfire prediction. Algorithms such as Random Forest, XGBoost, and Neural Networks have been trained on the large datasets to predict fire-prone areas based on historical data. Researchers have integrated AI models with real time sensor data to provide the accurate and timely fire predictions. The use of deep learning models, including (CNNs) and (RNNs), has enhanced the detection of fires from satellite images and meteorological reports. Wildfire prediction is a significant area of research in environmental science, meteorology, and artificial intelligence. As wildfires continue to cause devastating environmental and economic damage, researchers have focused on developing models that can analyse weather patterns, detect fire-prone areas, and predict fire occurrences with high accuracy. Several studies have demonstrated the effectiveness of ML and DL techniques in wildfire prediction:[1]Machine Learning-Based Fire Prediction :Researchers have utilized Decision Trees and Random Forest, and Support Vector Machines (SVM) to predict wildfire occurrences using meteorological data. Studies highlight the importance of temperature, humidity, wind speed, and precipitation as key variables influencing wildfire risk. Random Forest and Boosting models have shown high accuracy in classification tasks involving fire and no-fire zones.[2]Deep Learning and Remote Sensing in Fire Detection(CNNs) have been used for satellite image analysis, detecting fire outbreaks based on real time data.(RNNs) and Long Short-Term Memory (LSTM) models help predict fire

spread over time. NASA and other space agencies rely on geospatial satellite data to analyse fire-prone regions.[3]Feature Engineering and Data Analysis in Wildfire Prediction :Exploratory Data Analysis (EDA) has been widely applied to understand how different factors influence wildfire occurrences. A study showcased in the International Journal of Wildland Fire demonstrated that the Fire Weather Index (FWI) has a strong correlation (above 90%) with fire occurrences. Machine learning models have been trained on datasets from UCI, MODIS (Moderate Resolution Imaging Spectroradiometer), and regional fire databases.[4]Database Management for Large-Scale Fire Data :Relational databases (SQL) and NoSQL databases (MongoDB, Firebase) are used to store large wildfire datasets. MongoDB is often preferred due to its ability to handle unstructured and real-time data efficiently.

IV.RELATED WORK

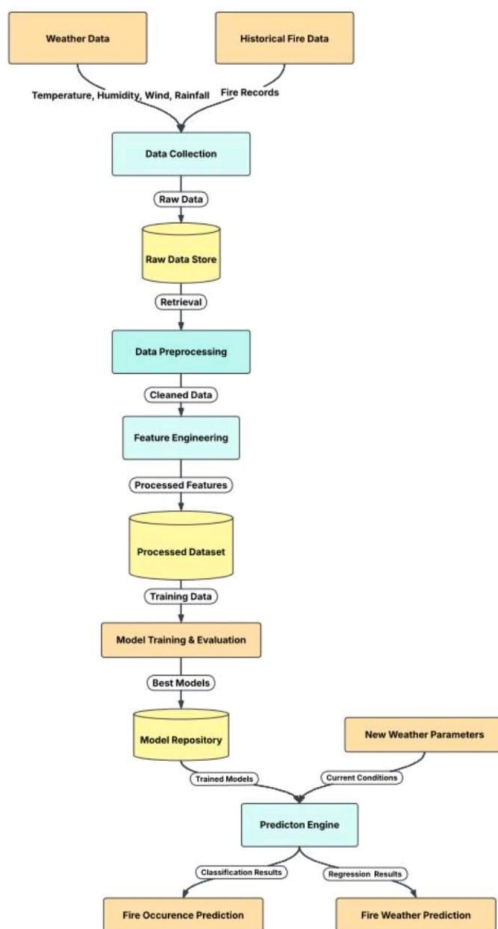
Wildfire prediction has been an important area for research field due to its devastating impact on the environment and human life. Numerous studies have been conducted using traditional statistical models, remote sensing techniques, and advanced machine learning algorithms to predict wildfire occurrences. One of the earliest approaches involved the use of meteorological data and fire factors, such as the (FWI), to assess wildfire risks. Studies showed that temperature and humidity and the wind speed, and precipitation played critical roles in fire outbreaks. With advancements in remote sensing and GIS (Geographic Information Systems), researchers started the usage of the satellite imagery to detect and monitor fires. NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite) provide real-time fire location data, which significantly improved prediction models. More recently, ML and DL have completely changes the wildfire prediction. Algorithms such as Random Forest, XGBoost, and Neural Networks have been trained on the large datasets to predict fire-prone areas based on historical data. Researchers have integrated AI models with real time sensor data to provide the accurate and timely fire predictions. The use of deep learning models, including (CNNs) and (RNNs), has enhanced the detection of fires from satellite images and meteorological reports. Wildfire prediction is a significant area of research in environmental science, meteorology, and artificial intelligence. As wildfires continue to cause devastating environmental and economic damage, researchers have focused on developing models that can analyse weather patterns, detect fire-prone areas, and predict fire occurrences with high accuracy

V.PROPOSED SYSTEM

Our wildfire prediction system is designed as a data-driven approach to predict and reduce wildfire threat using advanced machine learning techniques. By utilizing machine learning, the essential need for early warning systems and helping the authority intervene before the fire becomes severe. The approach efficiently does data collection, pre-processing, model training, making sure to continuously predict based on changing weather condition .Our Wildfire Prediction consists of key components that include:

1. A data pipeline that collects, cleans, and processes historical Wildfire and weather data.
2. Machine learning models design for classification (fire occurrence detection) and regression.
3. A cloud-based database for efficient data storage and retrieval.

By using these components, our system translates meteorological input into improved Wildfire prevention insights.



VI.IMPLEMENTATION

To implement the wildfire prediction model, a structured simulation setup was followed. The process involved multiple stages, including collection of the data preprocessing of data, (EDA), model training, evaluation, and final predictions. Technologies Used: Programming Language: Python Machine Learning Libraries: Scikit-learn, XGBoost, TensorFlow Data Analysis Tools: Pandas, NumPy, Matplotlib, Seaborn Database for Storage: MongoDB Steps Involved:[1]Data Collection: The dataset used for this project was sourced from the UCI Machine Learning Repository, specifically the Algerian Forest Fires dataset[2]. Data Preprocessing: The dataset was cleaned, missing values were handled, and categorical variables were encoded.[3]. Exploratory Data Analysis (EDA): Visualization techniques were applied to understand the weather conditions and fire occurrences relationship.4. Feature Selection: Important

features such as temperature, wind speed, relative humidity, and the Fire Weather Index (FWI) were identified.[5]. Model Selection: Various ML models were trained and tested for both classification (fire/no fire) and regression (FWI prediction) tasks[6]. Model Evaluation: Performance metrics like Accuracy, Precision, Recall, R^2 Score, and Mean Squared Error (MSE) were used to select the best-performing models.[7]. Final Model Optimization: Hyperparameter tuning using Grid Search CV and Stratified K Fold Cross-Validation was applied to enhance model performance.

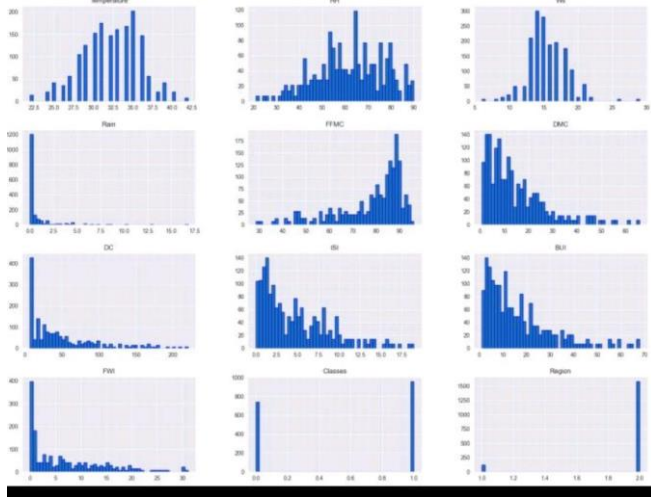
Hyperparameter tuning was performed using Randomized Research. Through extensive training and testing, the models demonstrated strong predictive capabilities, particularly Random Forest for classification and XGBoost for regression. The results indicate that machine learning can effectively predict wildfire risks based on weather conditions and historical fire data. This implementation can be further enhanced by integrating real-time weather API data and expanding the dataset with global wildfire records. Future work can focus on improving the model's ability to adapt to different environmental conditions and incorporating deep learning for more accurate image-based fire detection

VII.RESULT AND DISCUSSION

The proposed wildfire prediction model was developed using meteorological data from the UCI Machine Learning Repository, covering forest regions in Bejaia and Sidi Bel-Abbes (Algeria). The dataset included features such as temperature, relative humidity, wind speed, rainfall, and Fire Weather Index (FWI). After preprocessing (removing missing values, outlier treatment, and normalization), several machine learning algorithms were trained and evaluated for both classification and regression tasks .The Random Forest Classifier provided the best balance between bias and variance, demonstrating strong generalization capabilities. The high precision and recall values indicate reliable fire detection with minimal false positives or negatives .Regression algorithms including Linear Regression, Support Vector Regressor (SVR), and Random Forest Regressor were tested to forecast the Fire Weather Index (FWI).The experimental results confirm that ensemble models, particularly Random Forest, outperform single learners due to their robustness against overfitting and ability to handle multivariate, nonlinear data. The integration of data science and machine learning significantly enhances the accuracy and interpretability of wildfire prediction systems.

This study demonstrates that a data-driven approach can provide actionable insights for early warning, resource allocation, and disaster management. Future enhancements include the integration of real-time weather APIs, satellite imagery, and IoT-based sensor data to improve model adaptability and extend its application to larger and more diverse geographic areas.

Figure:



VIII. FUTURE SCOPE

Wildlife, human groups, and the environment are all severely impacted by wildfires. Leveraging data science and technological breakthroughs is compelling in improving the potency and precision of wildfire forecasts. This article highlights important upcoming advancements that have the potential to greatly enhance our existing wildfire forecast model.1. Connectivity to Windy. com's Real-Time Weather Data: Our wildfire prediction model's integration of real-time weather data from Windy.com is among its most promising enhancements. Important weather data that directly affects wildfire hazards, including temperature, precipitation, wind speed, and humidity, are provided by this site. By dynamically integrating various meteorological factors, the model can provide forecasts that are more precise and timely while also adjusting to changes in the environment as they happen.2. Creation of a Prediction System in Real Time: Our model currently uses static datasets, which means that it makes predictions based on historical data. We intend to create a real-time prediction system that is updated constantly in response to the most recent weather conditions in order to boost its effectiveness. The technology is able to produce immediate notifications for regions with a high risk of wildfires by automating the collection and processing of data. In addition to increasing prediction accuracy, this real-time approach will facilitate proactive disaster management, enabling officials to act swiftly and minimize any damage before fires escalate out of control.3. Putting Advanced Machine Learning Algorithms into Practice: Our goal is to incorporate cutting-edge machine learning procedures like Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks in order to notably improve prediction accuracy. Because of their expertise in examining time-series data, these algorithms are especially helpful in identifying trends in weather variations that fuel wildfire

breakouts.

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